

1-2 State the SI units and unit symbols for *force* and *work*, and define each unit.
1-3 State the SI units and unit symbols for *energy* and *power*, and define each unit.
1-4 List the names of the various metric prefixes and the corresponding symbols. Also, list the value represented by each prefix in scientific notation.
1-5 State the SI units and unit symbols for *electric current* and *charge*, and define each unit.
1-6 State the SI units and unit symbols for *electrical resistance* and *conductance*, and define each unit.
1-7 State the SI units and unit symbols for *magnetic flux* and *flux density*, and define each unit.
1-8 State the SI units and unit symbols for *inductance* and *capacitance*, and define each unit.
1-9 Name the two SI temperature scales, and identify the freezing and boiling temperatures of water for each scale.
1-10 List the various levels of measurement standards, and discuss the application of each classification.

PROBLEMS

1-1 Referring to the unit conversion factors in Appendix 1, perform the following conversions: (a) 113.5 miles to kilometers, (b) 50 miles per hour to kilometers per hour, and (c) 12 square feet to square centimeters.
1-2 Determine how long it takes light to travel to earth from a star 1 million miles away if the speed of light is 3×10^8 m/s.
1-3 The speed of sound in air is 345 m/s. Calculate the distance in miles from a thunderstorm when the thunder is heard 5 s after the lightning flash.
1-4 A 140 lb person has a height of 5 ft 7 in. Convert these measurements into kilograms and centimeters.
1-5 A bar magnet has a cross section of 0.75 in. x 0.75 in. and a flux density of 1200 lines per square inch. Calculate the total flux in webers.
1-6 Calculate the Celsius and Kelvin scale equivalents of 80°F.
1-7 A ¼ horsepower electric motor is operated 8 hours per day for 5 days every week. A

prompt

Here are the complete answers and step-by-step solutions for all the review questions and problems shown in your textbook page image (or let me know if you only needed specific section solutions instead).

Review Questions

1-1 List the three fundamental SI mechanical units and unit symbols, and discuss their origin.

> **Meter (m):** Unit of length. Originally defined as $1/10,000,000$ of the distance from the equator to the North Pole along the Paris meridian. (Currently defined by the distance light travels in a vacuum in $1/299,792,458$ seconds).
>
> **Kilogram (kg):** Unit of mass. Originally defined as the mass of 1 liter ($1000\ \text{cm}^3$) of pure water at 4°C. (Currently defined using Planck's constant).
>
> **Second (s):** Unit of time. Originally defined as $1/86,400$ of a mean solar day. (Currently defined by the transition frequency of cesium-133 atoms).

1-2 State the SI units and unit symbols for *force* and *work*, and define each unit.

> **Force:** Newton (N). $1\ \text{N}$ is the force required to accelerate a mass of $1\ \text{kg}$ at a rate of $1\ \text{m/s}^2$ ($1\ \text{N} = 1\ \text{kg} \cdot \text{m/s}^2$).
>
> **Work:** Joule (J). $1\ \text{J}$ is the work done when a force of $1\ \text{N}$ acts through a distance of $1\ \text{m}$ ($1\ \text{J} = 1\ \text{N} \cdot \text{m}$).

1-3 State the SI units and unit symbols for *energy* and *power*, and define each unit.

Energy: Joule (J). The capacity to do work; equal to the energy expended when applying $1\ \mathrm{N}$ over a displacement of $1\ \mathrm{m}$.

Power: Watt (W). $1\ \mathrm{W}$ is the rate of work done or energy transferred at $1\ \mathrm{J}$ per second ($1\ \mathrm{W} = 1\ \mathrm{J/s}$).

1-4 List the names of the various metric prefixes and the corresponding symbols. Also, list the value represented by each prefix in scientific notation.

Tera (T): 10^{12}

Giga (G): 10^{9}

Mega (M): 10^{6}

Kilo (k): 10^{3}

Hecto (h): 10^{2}

Deka (da): 10^{1}

Deci (d): 10^{-1}

Centi (c): 10^{-2}

Milli (m): 10^{-3}

Micro (μ): 10^{-6}

Nano (n): 10^{-9}

Pico (p): 10^{-12}

Femto (f): 10^{-15}

Atto (a): 10^{-18}

1-5 State the SI units and unit symbols for *electric current* and *charge*, and define each unit.

Electric Current: Ampere (A). The flow rate of electric charge equal to one Coulomb per second ($\mathrm{A} = \mathrm{C/s}$).

Electric Charge: Coulomb (C). The quantity of electricity transferred by a current of $1\ \mathrm{A}$ flowing for $1\ \mathrm{s}$ ($\mathrm{C} = \mathrm{A} \cdot \mathrm{s}$).

1-6 State the SI units and unit symbols for *electrical resistance* and *conductance*, and define each unit.

Electrical Resistance: Ohm (Ω). The resistance between two points when a constant potential difference of $1\ \mathrm{V}$ produces a current of $1\ \mathrm{A}$ ($1\ \Omega = 1\ \mathrm{V/A}$).

Electrical Conductance: Siemens (S). The reciprocal of resistance; $1\ \mathrm{S}$ is the conductance of a conductor with a resistance of $1\ \Omega$ ($1\ \mathrm{S} = 1/\Omega = 1\ \mathrm{A/V}$).

1-7 State the SI units and unit symbols for *magnetic flux* and *flux density*, and define each unit.

Magnetic Flux: Weber (Wb). $1\ \mathrm{Wb}$ is the flux that produces an electromotive force of $1\ \mathrm{V}$ in a single-turn circuit as it collapses to zero at a uniform rate in $1\ \mathrm{s}$ ($1\ \mathrm{Wb} = 1\ \mathrm{V \cdot s}$).

Flux Density: Tesla (T). A magnetic flux density of $1\ \mathrm{Wb}$ per square meter ($1\ \mathrm{T} = 1\ \mathrm{Wb/m^2}$).

1-8 State the SI units and unit symbols for *inductance* and *capacitance*, and define each unit.

Inductance: Henry (H). $1\ \mathrm{H}$ is the inductance of a circuit in which an electromotive force of $1\ \mathrm{V}$ is induced when the current changes at $1\ \mathrm{A/s}$ ($1\ \mathrm{H} = 1\ \mathrm{V \cdot s/A}$).

Capacitance: Farad (F). $1\ \mathrm{F}$ is the capacitance of a capacitor that holds a charge of $1\ \mathrm{C}$ under a potential difference of $1\ \mathrm{V}$ ($1\ \mathrm{F} = 1\ \mathrm{C/V}$).

1-9 Name the two SI temperature scales, and identify the freezing and boiling temperatures of water for each scale.

Celsius Scale (°C): Freezing point = 0°C, Boiling point = 100°C.

Kelvin Scale (K): Freezing point = 273.15 K, Boiling point = 373.15 K.

1-10 List the various levels of measurement standards, and discuss the application of each classification.

1. **International Standards:** Maintained by international organizations (e.g., BIPM) to provide a global baseline for standard evaluations.
2. **Primary Standards:** Maintained by national standard laboratories (e.g., NIST) to calibrate secondary standards and serve as absolute national references.
3. **Secondary Standards:** Maintained by industrial research laboratories to calibrate working standards for specific industries.
4. **Working Standards:** The primary measurement tools used for everyday calibration, testing, and production in laboratories and manufacturing plants.

Problems

1-1

(a) 6215 miles to kilometers:

$$\text{Distance} = 6215 \text{ mi} \times 1.60934 \text{ km/mi} = 10,002.05 \text{ km}$$

(b) 50 miles per hour to kilometers per hour:

$$\text{Speed} = 50 \text{ mph} \times 1.60934 \text{ km/mi} = 80.47 \text{ km/h}$$

(c) 12 square feet to square centimeters:

$$\text{Area} = 12 \text{ ft}^2 \times (30.48 \text{ cm/ft})^2 = 12 \times 929.03 \text{ cm}^2 = 11,148.36 \text{ cm}^2 \approx 1.115 \times 1$$

1-2

$$\text{Distance } d = 1,000,000 \text{ miles} \times 1609.344 \text{ m/mile} = 1.609344 \times 10^9 \text{ m}$$

$$\text{Speed of light } c = 3 \times 10^8 \text{ m/s}$$

$$\text{Time } t = \frac{d}{c} = \frac{1.609344 \times 10^9}{3 \times 10^8} = 5.36 \text{ seconds}$$

1-3

$$\text{Distance } d = \text{speed} \times \text{time} = 345 \text{ m/s} \times 5 \text{ s} = 1725 \text{ m}$$

$$\text{Distance in miles} = \frac{1725 \text{ m}}{1609.344 \text{ m/mile}} \approx 1.07 \text{ miles}$$

1-4

$$\text{Weight in kg} = 140 \text{ lb} \times 0.453592 \text{ kg/lb} \approx 63.5 \text{ kg}$$

$$\text{Height} = 5 \text{ ft } 7 \text{ in} = (5 \times 12) + 7 = 67 \text{ inches}$$

$$\text{Height in cm} = 67 \text{ in} \times 2.54 \text{ cm/in} \approx 170.18 \text{ cm}$$

1-5

$$\text{Area } A = 0.75 \text{ in.} \times 0.75 \text{ in.} = 0.5625 \text{ in.}^2$$

$$\text{Total flux in lines} = 1290 \text{ lines/in.}^2 \times 0.5625 \text{ in.}^2 = 725.625 \text{ lines}$$

Since $1 \text{ Weber} = 10^8$ lines:

$$\Phi = \frac{725.625}{10^8} = 7.256 \times 10^{-6} \text{ Wb} = 7.26 \ \mu\text{Wb}$$

1-6

$$\text{Celsius } T(^\circ\text{C}) = \frac{5}{9} \times (80 - 32) = \frac{5}{9} \times 48 = 26.67^\circ\text{C}$$

$$\text{Kelvin } T(\text{K}) = 26.67 + 273.15 = 299.82 \text{ K}$$

1-7

$$\text{Power } P = \frac{1}{4} \text{ hp} = 0.25 \times 746 \text{ W} = 186.5 \text{ W} = 0.1865 \text{ kW}$$

$$\text{Total Operating Hours in 1 Year} = 8 \text{ h/day} \times 5 \text{ days/wk} \times 52 \text{ wks/yr} = 2080 \text{ h}$$

$$\text{Energy Consumed} = 0.1865 \text{ kW} \times 2080 \text{ h} \approx 387.92 \text{ kWh}$$

1-8

In terms of base dimensions $[L]$ (length) and $[T]$ (time):

Area: $[L^2]$

Volume: $[L^3]$

Velocity: $[LT^{-1}]$

Acceleration: $[LT^{-2}]$

1-9

In terms of base dimensions $[M]$ (mass), $[L]$ (length), and $[T]$ (time):

Force: $[MLT^{-2}]$

Work: $[ML^2T^{-2}]$

Energy: $[ML^2T^{-2}]$

Power: $[ML^2T^{-3}]$

1-10

In terms of base dimensions $[M]$, $[L]$, $[T]$, and $[I]$ (electric current):

Charge: $[IT]$

Voltage: $[ML^2T^{-3}I^{-1}]$

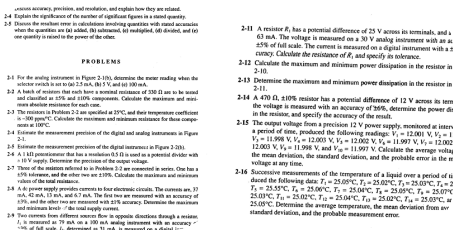
Resistance: $[ML^2T^{-3}I^{-2}]$

1-11

In terms of base dimensions $[M]$, $[L]$, $[T]$, and $[I]$ (electric current):

Capacitance: $[M^{-1}L^{-2}T^4I^2]$

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Here are the complete answers and step-by-step solutions for the Chapter 2 Review Questions and Problems provided in the images (or let me know if you only needed solutions for a specific problem or section instead). PNG + 1

Review Questions

2-1. Explain gross errors and systematic errors. Give examples of each.

Gross Errors: Errors caused by human oversight, carelessness, reading instrument scales incorrectly, recording data faulty, or improper circuit connection. PNG

> *Example:* Recording a meter reading as $45\ \mathrm{mA}$ when it actually indicates $35\ \mathrm{mA}$, or misplacing a decimal point in data entry. PNG

Systematic Errors: Errors that consistently reoccur under the same measurement conditions due to equipment flaws, environmental influences, or observational bias. PNG

> *Example:* A meter reading $0.2\ \mathrm{V}$ when zero voltage is applied due to uncalibrated mechanical zero settings, or a resistor whose value shifts predictably due to ambient temperature rise. PNG

2-2. Define absolute errors and relative errors.

Absolute Error (e): The numerical difference between the measured value (Y_n) and the true/expected value (X_n): PNG

$$e = Y_n - X_n$$

Relative Error (e_r or $\%$ Error): The ratio of the absolute error to the true or expected value, typically expressed as a percentage: PNG

$$e_r = \frac{Y_n - X_n}{X_n} = \frac{e}{X_n}$$

$$\%\ \text{Error} = \left(\frac{Y_n - X_n}{X_n}\right) \times 100\%$$

2-3. Discuss accuracy, precision, and resolution, and explain how they are related.

Accuracy: How close a measured value is to the true or accepted value. PNG

Precision: The degree of consistency or repeatability among successive measurements of the same quantity under identical conditions. PNG

Resolution: The smallest change or increment in the measured quantity that an instrument can detect and display. PNG

Relationship: Precision sets the upper boundary for accuracy (a measurement cannot be accurate beyond its precision limits), while resolution defines the smallest detectable unit that contributes to precision and accuracy. PNG

2-4. Explain the significance of the number of significant figures in a stated quantity.

Significance: The number of significant figures indicates the precision and uncertainty of a measurement. It reflects both the magnitude of the measured value and the confidence in the measurement—the last digit represents an estimated/uncertain digit, while all preceding digits are known with certainty. PNG + 1

2-5. Discuss the resultant error in calculations involving quantities with stated accuracies

:

(a) Added: The maximum absolute error is the sum of the maximum absolute errors of individual quantities:

PNG

$$e_{\text{total}} = \pm(e_1 + e_2 + \dots)$$

(b) Subtracted: The maximum absolute error is the sum of the absolute errors of individual quantities:

PNG

$$e_{\text{total}} = \pm(e_1 + e_2 + \dots)$$

(c) Multiplied: The relative (percentage) error of the result is approximately equal to the sum of the relative (percentage) errors of the individual factors:

PNG

$$\%e_{\text{total}} \approx \pm(\%e_1 + \%e_2 + \dots)$$

(d) Divided: The relative (percentage) error of the result is approximately equal to the sum of the relative (percentage) errors of the numerator and denominator:

PNG

$$\%e_{\text{total}} \approx \pm(\%e_1 + \%e_2 + \dots)$$

(e) Quantity Raised to a Power (A^n): The relative percentage error of A^n is n times the relative error of A:

PNG

$$\%e_{A^n} = \pm(n \times \%e_A)$$

Problems

Note on Problems 2-1, 2-4, and 2-5: *These three problems require Figure 2–1 and Figure 2–2(b), which are not present on the uploaded textbook pages. Solutions for all remaining fully specified problems are provided below.*

PNG + 2

Problem 2-2

Given $R = 330\ \Omega$ nominal. PNG

For $\pm 5\%$ components:

PNG

$$\text{Tolerance} = 330\ \Omega \times 0.05 = 16.5\ \Omega$$

$$\text{Minimum Resistance} = 330 - 16.5 = \mathbf{313.5\ \Omega}$$

$$\text{Maximum Resistance} = 330 + 16.5 = \mathbf{346.5\ \Omega}$$

For $\pm 10\%$ components:

PNG

$$\text{Tolerance} = 330\ \Omega \times 0.10 = 33\ \Omega$$

$$\text{Minimum Resistance} = 330 - 33 = \mathbf{297\ \Omega}$$

$$\text{Maximum Resistance} = 330 + 33 = \mathbf{363\ \Omega}$$

Problem 2-3

Given $R_{25^\circ\text{C}} = 330\ \Omega$, Temperature Coefficient $\alpha = -300\ \text{ppm}/^\circ\text{C} = -300 \times 10^{-6}/^\circ\text{C}$, $\Delta T = 100^\circ\text{C} - 25^\circ\text{C} = 75^\circ\text{C}$. PNG

Fractional shift in resistance:

$$\frac{\Delta R}{R} = (-300 \times 10^{-6}) \times 75 = -0.0225 \quad (-2.25\%)$$

For $\pm 5\%$ components at 100°C:

$$\text{Minimum Resistance} = 313.5 \times (1 - 0.0225) = \mathbf{306.45\ \Omega}$$

$$\text{Maximum Resistance} = 346.5 \times (1 - 0.0225) = \mathbf{338.70\ \Omega}$$

For $\pm 10\%$ components at 100°C:

$$\text{Minimum Resistance} = 297 \times (1 - 0.0225) = \mathbf{290.32\ \Omega}$$

$$\text{Maximum Resistance} = 363 \times (1 - 0.0225) = \mathbf{354.83\ \Omega}$$

Problem 2-6

Given $R = 1\ \text{k}\Omega = 1000\ \Omega$, resolution $\Delta R = 0.5\ \Omega$, supply voltage $V_s = 10\ \text{V}$. PNG

Output voltage resolution step:

$$\Delta V_{\text{out}} = \left(\frac{0.5\ \Omega}{1000\ \Omega} \right) \times 10\ \text{V} = 0.005\ \text{V} = 5\ \text{mV}$$

Precision of output voltage: $\pm \mathbf{2.5}\ \text{mV}$ (or resolution step of $\mathbf{5}\ \text{mV}$).

Problem 2-7

Three $330\ \Omega$ nominal resistors in series: one at $\pm 5\%$, two at $\pm 10\%$. PNG

$$\text{Minimum Total Resistance} = 313.5 + 297 + 297 = \mathbf{907.5\ \Omega}$$

$$\text{Maximum Total Resistance} = 346.5 + 363 + 363 = \mathbf{1072.5\ \Omega}$$

Problem 2-8

Given currents: PNG

$$I_1 = 37\ \text{mA} \pm 3\% \implies e_1 = \pm 1.11\ \text{mA}$$

PNG

$$I_2 = 42\ \text{mA} \pm 3\% \implies e_2 = \pm 1.26\ \text{mA}$$

PNG

$$I_3 = 13\ \text{mA} \pm 1\% \implies e_3 = \pm 0.13\ \text{mA}$$

PNG

$$I_4 = 6.7\ \text{mA} \pm 1\% \implies e_4 = \pm 0.067\ \text{mA}$$

PNG

Nominal total current:

$$I_{\text{total}} = 37 + 42 + 13 + 6.7 = 98.7\ \text{mA}$$

Total absolute error:

$$e_{\text{total}} = \pm(1.11 + 1.26 + 0.13 + 0.067) = \pm 2.567\ \text{mA}$$

$$\text{Minimum Total Current} = 98.7 - 2.567 = \mathbf{96.133}\ \text{mA}$$

$$\text{Maximum Total Current} = 98.7 + 2.567 = \mathbf{101.267}\ \text{mA}$$

Problem 2-9

Given $I_1 = 79\ \text{mA}$ measured on $100\ \text{mA}$ analog scale ($\pm 3\%$ of full scale $\implies e_1 = \pm 3\ \text{mA}$) and $I_2 = 31\ \text{mA}$ measured digitally ($\pm 100\ \mu\text{A} = \pm 0.1\ \text{mA}$). PNG

Flowing in opposite directions:

$$I_{\text{net}} = I_1 - I_2 = 79 - 31 = 48\ \text{mA}$$

$$e_{\text{total}} = \pm(3 + 0.1) = \pm 3.1\ \text{mA}$$

$$\text{Minimum Current in } R_1 = 48 - 3.1 = \mathbf{44.9}\ \text{mA}$$

$$\text{Maximum Current in } R_1 = 48 + 3.1 = \mathbf{51.1}\ \text{mA}$$

Problem 2-10

Given $R = 470\ \Omega \pm 5\%$, $V_1 = 12\ \text{V} \pm 0.5\ \text{V}$, $V_2 = 5\ \text{V} \pm 2\%$ ($5\ \text{V} \pm 0.1\ \text{V}$). PNG

Voltage across resistor:

$$V = V_1 - V_2 = 12 - 5 = 7\ \text{V} \pm 0.6\ \text{V} \quad (\pm 8.57\%)$$

Nominal current:

$$I = \frac{7\ \text{V}}{470\ \Omega} = \mathbf{14.89}\ \text{mA}$$

Accuracy of current:

$$\%e_I = \%e_V + \%e_R = 8.57\% + 5\% = \pm\mathbf{13.57\%} \quad (\pm 2.02\ \text{mA})$$

(Extreme range: $I_{\min} = \mathbf{12.97}\ \text{mA}$, $I_{\max} = \mathbf{17.02}\ \text{mA}$)

Problem 2-11

Given $V = 25\ \text{V}$ measured on $30\ \text{V}$ analog scale ($\pm 5\%$ full scale $\implies \Delta V = \pm 1.5\ \text{V} \implies \pm 6\%$) and $I = 63\ \text{mA} \pm 1\ \text{mA}$ ($\pm 1.59\%$). PNG

Resistance of R_1:

$$R_1 = \frac{25\ \text{V}}{0.063\ \text{A}} = \mathbf{396.83\ \Omega}$$

Tolerance of R_1:

$$\%e_R = 6\% + 1.59\% = \pm\mathbf{7.59\%} \quad (\pm 30.11\ \Omega)$$

(Extreme range: $R_{\text{min}} = \mathbf{367.19\ \Omega}$, $R_{\text{max}} = \mathbf{427.42\ \Omega}$)

Problem 2-12

Using parameters from Problem 2-10 ($V_{\text{min}} = 6.4\ \text{V}$, $V_{\text{max}} = 7.6\ \text{V}$, $R_{\text{min}} = 446.5\ \Omega$, $R_{\text{max}} = 493.5\ \Omega$): PNG + 1

$$\text{Nominal Power } P_{\text{nom}} = \frac{7^2}{470} = \mathbf{104.26}\ \text{mW}$$

$$\text{Minimum Power } P_{\text{min}} = \frac{6.4^2}{493.5} = \mathbf{83.00}\ \text{mW}$$

$$\text{Maximum Power } P_{\text{max}} = \frac{7.6^2}{446.5} = \mathbf{129.36}\ \text{mW}$$

Problem 2-13

Using parameters from Problem 2-11 ($V = 25\ \text{V} \pm 1.5\ \text{V}$, $I = 63\ \text{mA} \pm 1\ \text{mA}$): PNG

$$\text{Nominal Power } P_{\text{nom}} = 25\ \text{V} \times 63\ \text{mA} = \mathbf{1.575}\ \text{W}\ (1575\ \text{mW})$$

$$\text{Minimum Power } P_{\text{min}} = 23.5\ \text{V} \times 62\ \text{mA} = \mathbf{1.457}\ \text{W}\ (1457\ \text{mW})$$

$$\text{Maximum Power } P_{\text{max}} = 26.5\ \text{V} \times 64\ \text{mA} = \mathbf{1.696}\ \text{W}\ (1696\ \text{mW})$$

Problem 2-14

Given $R = 470\ \Omega \pm 10\%, V = 12\ \text{V} \pm 6\%$. PNG

$$\text{Power Dissipation } P = \frac{12^2}{470} = \mathbf{0.3064}\ \text{W}\ (306.4\ \text{mW})$$

$$\text{Accuracy of Power } \%e_P = 2(\%e_V) + \%e_R = 2(6\%) + 10\% = \pm\mathbf{22\%} \quad (\pm 67.4\ \text{m}$$

Problem 2-15

Data:
$12.001, 11.999, 11.998, 12.003, 12.002, 11.997, 12.002, 12.003, 11.998, 11.997\ \text{V}$.
PNG

Average Voltage ($\bar{V}$):

$$\bar{V} = \frac{\sum V}{10} = \mathbf{12.0000}\ \text{V}$$

Mean Deviation ($\bar{D}$):

$$\bar{D} = \frac{\sum |V - \bar{V}|}{10} = \frac{0.022}{10} = \mathbf{0.0022}\ \text{V}\ (2.20\ \text{mV})$$

Standard Deviation (s):

$$s = \sqrt{\frac{\sum (V - \bar{V})^2}{n-1}} = \sqrt{\frac{0.000054}{9}} = \mathbf{0.00245}\ \text{V}\ (2.45\ \text{mV})$$

Probable Error (r):

$$r = 0.6745 \times s = 0.6745 \times 0.00245 = \mathbf{0.00165}\ \text{V}\ (1.65\ \text{mV})$$

Problem 2-16

PNG + 4

Data:
$25.05, 25.02, 25.03, 25.07, 25.55, 25.06, 25.04, 25.05, 25.07, 25.03, 25.02, 25.04, 25.0$
. PNG

Average Temperature ($\bar{T}$):

$$\bar{T} = \frac{\sum T}{15} = \mathbf{25.0753}^\circ\text{C}$$

Mean Deviation ($\bar{D}$):

$$\bar{D} = \frac{\sum |T - \bar{T}|}{15} = \mathbf{0.0633}^\circ\text{C}$$

Standard Deviation (s):

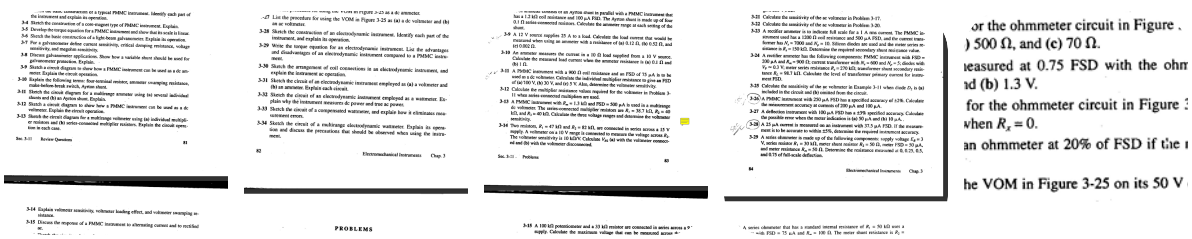
$$s = \sqrt{\frac{\sum (T - \bar{T})^2}{n - 1}} = \mathbf{0.1324}^\circ\text{C}$$

Probable Measurement Error (r):

$$r = 0.6745 \times s = 0.6745 \times 0.1324 = \mathbf{0.0893}^\circ\text{C}$$

(Note: If $T_5 = 25.05^\circ\text{C}$ without a textbook typo: $\bar{T} = \mathbf{25.0420}^\circ\text{C}$, $\bar{D} = \mathbf{0.0141}^\circ\text{C}$, $s = \mathbf{0.0170}^\circ\text{C}$, $r = \mathbf{0.0115}^\circ\text{C}$).

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Here is a comprehensive set of step-by-step solutions for the Chapter 3 Review Questions and Numerical Problems provided in your textbook pages (or let me know if you only needed answers for specific questions). PNG + 4

Part 1: Review Questions Key Concepts

3-1 Moving System Forces: Deflecting torque (moves pointer proportional to quantity), Controlling torque (opposes motion and returns pointer to zero via spiral springs/taut-band), and Damping torque (prevents oscillation using eddy currents or air friction). PNG + 2

3-2 Suspensions: Jeweled-bearing uses pivot and jewel bearings (durable but has micro-friction), whereas Taut-band suspension uses stretched metallic ribbon (frictionless, highly sensitive, resistant to shock). PNG + 1

3-3 & 3-5 PMMC Principles & Linear Scale: Deflecting torque is given by $T_d = B \cdot A \cdot N \cdot I$. Since magnetic flux density B, area A, and turn count N are constant, $T_d \propto I$. Controlling torque from springs is $T_c = K \cdot \theta$. At equilibrium, $T_d = T_c \implies \theta \propto I$, producing a perfectly **linear scale**. PNG + 4

3-7 Galvanometer Terms:

Current Sensitivity (S_I): Deflection per unit current ($\mathrm{mm}/\mu\mathrm{A}$ or $\mu\mathrm{A}/\mathrm{mm}$). PNG

Critical Damping Resistance (R_c): Total circuit resistance required for critical damping. PNG

Voltage Sensitivity (S_V): $S_V = S_I \cdot R_c$. PNG

Megohm Sensitivity (S_M): Resistance in $\mathrm{M}\Omega$ required in series with $1\ \mathrm{V}$ to produce $1\ \mathrm{mm}$ deflection. PNG

3-10 Ammeter Concepts:

Four-terminal resistor: Low-resistance shunt with separate current and potential terminals to eliminate contact/lead resistance errors. PNG

Swamping resistance: Temperature-stable resistor (Manganin) connected in series with meter coil to minimize copper resistance drift with temperature. PNG

Ayrton Shunt: Multi-range shunt circuit that keeps the meter protected and prevents open-circuit conditions during switching. PNG

3-14 Voltmeter Concepts:

Sensitivity (S): Internal resistance per volt ($S = 1/I_{\text{FSD}}$ $[\Omega/\text{V}]$). PNG

Loading Effect: The error introduced when a voltmeter draws current and drops the effective resistance across the component being measured. PNG

3-20 Parallax Error: Error caused by looking at the pointer from an angle. It is minimized by aligning the pointer image with its reflection in a mirrored scale. PNG + 1

Part 2: Numerical Problems Solutions

Problem 3-1

Given $N = 300$, $B = 0.15\ \text{T}$, $D = 1.25\ \text{cm} = 0.0125\ \text{m}$, $l = 2\ \text{cm} = 0.02\ \text{m}$, $I = 500\ \mu\text{A} = 500 \times 10^{-6}\ \text{A}$.
PNG

$$\text{Area } A = D \times l = 0.0125\ \text{m} \times 0.02\ \text{m} = 0.00025\ \text{m}^2$$

$$\text{Torque } T = N \cdot B \cdot A \cdot I = 300 \times 0.15 \times 0.00025 \times (500 \times 10^{-6})$$

$$T = \mathbf{5.625 \times 10^{-6}}\ \text{N} \cdot \text{m} \quad (\mathbf{5.625}\ \mu\text{N} \cdot \text{m})$$

Problem 3-2

Given $B = 0.12\ \text{T}$, $D = 1.5\ \text{cm} = 0.015\ \text{m}$, $l = 2.25\ \text{cm} = 0.0225\ \text{m}$, $T = 4.5\ \mu\text{N}\cdot\text{m}$, $I = 100\ \mu\text{A}$.

PNG

$$A = 0.015 \times 0.0225 = 3.375 \times 10^{-4}\ \text{m}^2$$

$$N = \frac{T}{B \cdot A \cdot I} = \frac{4.5 \times 10^{-6}}{0.12 \times (3.375 \times 10^{-4}) \times (100 \times 10^{-6})} = \mathbf{1111}\ \text{turns}$$

Problem 3-3

Given $S_I = 500\ \text{nA/mm} = 0.5\ \mu\text{A/mm}$, $R_c = 3\ \text{k}\Omega = 3000\ \Omega$. PNG

Voltage Sensitivity (S_V):

$$S_V = S_I \times R_c = (500 \times 10^{-9}\ \text{A/mm}) \times 3000\ \Omega = \mathbf{1.5}\ \text{mV/mm}$$

Megohm Sensitivity (S_M):

$$S_M = \frac{1}{S_I \text{ in } \mu\text{A/mm}} = \frac{1}{0.5} = \mathbf{2.0}\ \text{M}\mathbf{\Omega}$$

Problem 3-4

Given $S_V = 300\ \mu\text{V/mm}$, $S_M = 1.5\ \text{M}\Omega$.

PNG + 1

$$S_I = \frac{1}{S_M} = \frac{1}{1.5\ \text{M}\Omega} = \frac{2}{3}\ \mu\text{A/mm} \approx 0.667\ \mu\text{A/mm}$$

$$\text{Critical Damping Resistance } R_c = \frac{S_V}{S_I} = \frac{300\ \mu\text{V/mm}}{(2/3)\ \mu\text{A/mm}} = \mathbf{450\ \Omega}$$

Problem 3-5

Given deflection $d = 5\text{ cm} = 50\text{ mm}$ at $I = 20\ \mu\text{A}$. PNG

Current Sensitivity (S_I):

$$S_I = \frac{I}{d} = \frac{20\ \mu\text{A}}{50\text{ mm}} = \mathbf{0.4}\ \mu\text{A/mm} \quad (\text{or } \mathbf{2.5}\ \text{mm}/\mu\text{A})$$

Megohm Sensitivity (S_M):

$$S_M = \frac{1}{0.4} = \mathbf{2.5}\ \text{M}\mathbf{\Omega}$$

Problem 3-6

Given $R_m = 750\ \Omega, I_m = 500\ \mu\text{A} = 0.5\text{ mA}$. PNG

(a) For $I = 50\text{ mA}$ FSD:

$$R_{sh} = \frac{I_m R_m}{I - I_m} = \frac{0.5\text{ mA} \times 750\ \Omega}{50\text{ mA} - 0.5\text{ mA}} = \frac{375}{49.5} = \mathbf{7.58\ \Omega}$$

(b) For $I = 30\text{ mA}$ FSD:

$$R_{sh} = \frac{0.5\text{ mA} \times 750\ \Omega}{30\text{ mA} - 0.5\text{ mA}} = \frac{375}{29.5} = \mathbf{12.71\ \Omega}$$

Problem 3-7

Given $R_{sh} = 133.3\ \Omega$, $R_m = 1.2\ \text{k}\Omega = 1200\ \Omega$, $I_m = 30\ \mu\text{A}$ at FSD.

PNG

$$I = I_m \left(1 + \frac{R_m}{R_{sh}}\right) = 30\ \mu\text{A} \left(1 + \frac{1200}{133.3}\right)$$

At FSD: $I = \mathbf{300}\ \mu\text{A}\ (\mathbf{0.300}\ \text{mA})$

At 0.5 FSD: $I = 0.5 \times 300\ \mu\text{A} = \mathbf{150}\ \mu\text{A}\ (\mathbf{0.150}\ \text{mA})$

At 0.33 FSD: $I = 0.33 \times 300\ \mu\text{A} = \mathbf{99}\ \mu\text{A}\ (\mathbf{0.099}\ \text{mA})$

Problem 3-9

Source $V = 12\ \text{V}$, $I = 25\ \text{A} \implies R_L = \frac{12\ \text{V}}{25\ \text{A}} = 0.48\ \Omega$.

PNG

$$I_{\text{measured}} = \frac{12\ \text{V}}{0.48\ \Omega + R_a}$$

(a) $R_a = 0.12\ \Omega$: $I_m = \frac{12}{0.48+0.12} = \mathbf{20.00}\ \text{A}$

(b) $R_a = 0.52\ \Omega$: $I_m = \frac{12}{0.48+0.52} = \mathbf{12.00}\ \text{A}$

(c) $R_a = 0.002\ \Omega$: $I_m = \frac{12}{0.48+0.002} = \mathbf{24.90}\ \text{A}$

Problem 3-10

Source $V = 10\ \text{V}$, $R_L = 10\ \Omega$. PNG

(a) $R_a = 0.1\ \Omega$: $I_m = \frac{10}{10+0.1} = \mathbf{0.990}\ \text{A}$

(b) $R_a = 1\ \Omega$: $I_m = \frac{10}{10+1} = \mathbf{0.909}\ \text{A}$

Problem 3-11

Given $R_m = 900\ \Omega$, $I_m = 75\ \mu\text{A}$. PNG

Voltmeter Sensitivity:

$$S = \frac{1}{I_m} = \frac{1}{75 \times 10^{-6}\ \text{A}} = \mathbf{13.33\ k\Omega/V}$$

Individual Multiplier Resistors ($R_{\text{mult}} = V \cdot S - R_m$):

(a) 100 V range: $R_{m1} = (100 \times 13.333\ \text{k}\Omega/\text{V}) - 0.9\ \text{k}\Omega = \mathbf{1332.43\ k\Omega}\ (1.332\ \text{M}\Omega)$

(b) 30 V range: $R_{m2} = (30 \times 13.333\ \text{k}\Omega/\text{V}) - 0.9\ \text{k}\Omega = \mathbf{399.10\ k\Omega}$

(c) 5 V range: $R_{m3} = (5 \times 13.333\ \text{k}\Omega/\text{V}) - 0.9\ \text{k}\Omega = \mathbf{65.77\ k\Omega}$

Problem 3-12

Series-connected multipliers using Problem 3-11 components: PNG

R_3 (for 5 V range): $R_3 = \mathbf{65.77\ k\Omega}$

R_2 (for 30 V range): $R_2 = \frac{30\ \text{V} - 5\ \text{V}}{75\ \mu\text{A}} = \mathbf{333.33\ k\Omega}$

R_1 (for 100 V range): $R_1 = \frac{100\ \text{V} - 30\ \text{V}}{75\ \mu\text{A}} = \mathbf{933.33\ k\Omega}$

Problem 3-13

Given $R_m = 1.3\ \text{k}\Omega$, $I_m = 500\ \mu\text{A}$, $R_1 = 38.7\ \text{k}\Omega$, $R_2 = 40\ \text{k}\Omega$, $R_3 = 40\ \text{k}\Omega$. PNG

Sensitivity:

$$S = \frac{1}{500\ \mu\text{A}} = \mathbf{2\ k\Omega/V}$$

Voltage Ranges:

Range 1: $V_1 = I_m(R_m + R_1) = 0.5\ \text{mA} \times (1.3 + 38.7)\ \text{k}\Omega = \mathbf{20\ V}$

Range 2: $V_2 = I_m(R_m + R_1 + R_2) = 0.5\ \text{mA} \times (40 + 40)\ \text{k}\Omega = \mathbf{40}\ \text{V}$

Range 3: $V_3 = I_m(R_m + R_1 + R_2 + R_3) = 0.5\ \text{mA} \times (80 + 40)\ \text{k}\Omega = \mathbf{60}\ \text{V}$

Problem 3-14

Given $R_1 = 47\ \text{k}\Omega$, $R_2 = 82\ \text{k}\Omega$, $V_s = 15\ \text{V}$, Range $= 10\ \text{V}$, $S = 10\ \text{k}\Omega/\text{V} \implies R_v = 100\ \text{k}\Omega$. PNG

(a) Voltmeter connected across R_2:

$$R_p = R_2 \parallel R_v = \frac{82 \times 100}{82 + 100} = 45.055\ \text{k}\Omega$$

$$V_{R2} = 15\ \text{V} \times \frac{45.055}{47 + 45.055} = \mathbf{7.34}\ \text{V}$$

(b) Voltmeter disconnected:

$$V_{R2} = 15\ \text{V} \times \frac{82}{47 + 82} = \mathbf{9.54}\ \text{V}$$

Problem 3-15

Given $R_{\text{pot}} = 100\ \text{k}\Omega$, $R = 33\ \text{k}\Omega$, $V_s = 9\ \text{V}$. PNG

(a) $S = 20\ \text{k}\Omega/\text{V}$, Range $= 15\ \text{V} \implies R_v = 300\ \text{k}\Omega$:

$$R_p = 100 \parallel 300 = 75\ \text{k}\Omega \implies V_{\max} = 9\ \text{V} \times \frac{75}{75 + 33} = \mathbf{6.25}\ \text{V}$$

(b) $S = 100\ \text{k}\Omega/\text{V}$, Range $= 10\ \text{V} \implies R_v = 1000\ \text{k}\Omega$:

$$R_p = 100 \parallel 1000 = 90.91\ \text{k}\Omega \implies V_{\max} = 9\ \text{V} \times \frac{90.91}{90.91 + 33} = \mathbf{6.60}\ \text{V}$$

Problem 3-16

Given $R_1 = 70\ \text{k}\Omega$, $R_2 = 50\ \text{k}\Omega$, $V_s = 12\ \text{V}$, Range $= 5\ \text{V}$. PNG

(a) Voltmeter disconnected:

$$V_{R2} = 12\ \text{V} \times \frac{50}{120} = \mathbf{5.00}\ \text{V}$$

(b) $S = 20\ \text{k}\Omega/\text{V} \implies R_v = 100\ \text{k}\Omega$:

$$R_p = 50 \parallel 100 = 33.33\ \text{k}\Omega \implies V_{R2} = 12\ \text{V} \times \frac{33.33}{70 + 33.33} = \mathbf{3.87}\ \text{V}$$

(c) $S = 200\ \text{k}\Omega/\text{V} \implies R_v = 1000\ \text{k}\Omega$:

$$R_p = 50 \parallel 1000 = 47.62\ \text{k}\Omega \implies V_{R2} = 12\ \text{V} \times \frac{47.62}{70 + 47.62} = \mathbf{4.86}\ \text{V}$$

Problem 3-26

Given $I_{\text{FSD}} = 250\ \mu\text{A}$, accuracy $\pm 2\% \implies \text{Error} = \pm 5\ \mu\text{A}$. PNG

At $200\ \mu\text{A}$: Accuracy $= \pm\left(\frac{5\ \mu\text{A}}{200\ \mu\text{A}}\right) \times 100\% = \pm\mathbf{2.5\%}$

At $100\ \mu\text{A}$: Accuracy $= \pm\left(\frac{5\ \mu\text{A}}{100\ \mu\text{A}}\right) \times 100\% = \pm\mathbf{5.0\%}$

Problem 3-27

Given $I_{\text{FSD}} = 100\ \mu\text{A}$, specified accuracy $\pm 3\% \implies \text{Error} = \pm 3\ \mu\text{A}$. PNG

(a) At $50\ \mu\text{A}$ indication: Error $= \pm\left(\frac{3\ \mu\text{A}}{50\ \mu\text{A}}\right) \times 100\% = \pm\mathbf{6.0\%}$

(b) At $10\ \mu\text{A}$ indication: Error $= \pm\left(\frac{3\ \mu\text{A}}{10\ \mu\text{A}}\right) \times 100\% = \pm\mathbf{30.0\%}$

Problem 3-28

Given $I = 25\ \mu\text{A}$, $I_{\text{FSD}} = 37.5\ \mu\text{A}$, target measurement accuracy $= \pm 5\%$.

PNG

$$\text{Max Allowable Error} = 25\ \mu\text{A} \times 0.05 = \pm 1.25\ \mu\text{A}$$

$$\text{Required Instrument Accuracy at FSD} = \pm\left(\frac{1.25\ \mu\text{A}}{37.5\ \mu\text{A}}\right) \times 100\% = \pm\mathbf{3.33\%}$$

Problem 3-29

Given $E_b = 3\ \text{V}$, $R_1 = 30\ \text{k}\Omega$, $R_2 = 50\ \Omega$, $R_m = 50\ \Omega$, $I_{\text{FSD}} = 50\ \mu\text{A}$.

PNG

$$\text{Internal Resistance } R_{\text{in}} = R_1 + (R_2 \parallel R_m) = 30,000 + 25 = 30.025\ \text{k}\Omega$$

At 0 FSD: $R_x = \boldsymbol{\infty}$

At 0.25 FSD: $R_x = 3 \times R_{\text{in}} = \mathbf{90.08\ k\Omega}$

At 0.50 FSD: $R_x = R_{\text{in}} = \mathbf{30.03\ k\Omega}$

At 0.75 FSD: $R_x = \frac{R_{\text{in}}}{3} = \mathbf{10.01\ k\Omega}$

At 1.00 FSD: $R_x = \mathbf{0\ \Omega}$

Problem 3-30

Given $R_1 = 50\ \text{k}\Omega$, $I_{\text{FSD}} = 75\ \mu\text{A}$, $R_m = 100\ \Omega$, $R_2 = 300\ \Omega$, $E_b = 5\ \text{V}$.

PNG

$$R_{\text{in}} = 50,000 + (300 \parallel 100) = 50.075\ \text{k}\Omega$$

At 0 FSD: $R_x = \boldsymbol{\infty}$

At 25% FSD: $R_x = 3 \times 50.075\ \text{k}\Omega = \mathbf{150.23\ \text{k}\Omega}$

At 50% FSD: $R_x = \mathbf{50.08\ \text{k}\Omega}$

At 75% FSD: $R_x = \frac{50.075\ \text{k}\Omega}{3} = \mathbf{16.69\ \text{k}\Omega}$

At 100% FSD: $R_x = \mathbf{0\ \Omega}$

Problem 3-32

Meter specified accuracy $= \pm 2\%$ at FSD. PNG

At 0.5 FSD ($s = 0.5$):

$$\%e_{Rx} = \frac{\%\text{Accuracy}}{s(1-s)} = \frac{2\%}{0.5(1-0.5)} = \pm\mathbf{8.0\%}$$

At 0.75 FSD ($s = 0.75$):

$$\%e_{Rx} = \frac{2\%}{0.75(1-0.75)} = \frac{2\%}{0.1875} = \pm\mathbf{10.67\%}$$

Problem 3-37

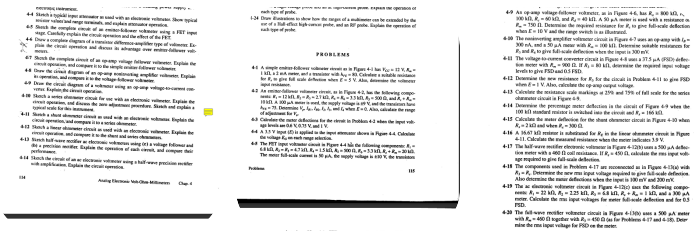
PNG + 4

Meter accuracy $= \pm 1\%$ at FSD, evaluated at 20% FSD ($s = 0.2$).

PNG

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$$\%e_{Rx} = \frac{1\%}{0.2 \times (1-0.2)} = \frac{1\%}{0.16} = \pm\mathbf{6.25\%}$$



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Here are the complete answers and step-by-step solutions for Chapter 4 Review Questions and Problems provided in your textbook pages (or let me know if you only needed specific section solutions instead). PNG + 2

Review Questions

4-1. Sketch the circuit of a simple transistor emitter-follower voltmeter. Explain the circuit operation, and compare it to a nonelectronic voltmeter.

Circuit & Operation: Consists of a BJT connected in an emitter-follower (common collector) configuration. The input voltage E is applied to the base, and a PMMC meter in series with a multiplier resistor R_s is connected in the emitter branch. The emitter voltage follows the input voltage ($V_E \approx E - V_{BE}$). Since current gain h_{FE} is high, the base input current is extremely small ($I_B \approx I_E/h_{FE}$). PNG + 3

Comparison: Unlike a conventional nonelectronic PMMC voltmeter (whose input resistance depends directly on meter sensitivity, e.g., $20\ \text{k}\Omega/\text{V}$), the electronic emitter-follower voltmeter offers a significantly higher input resistance ($R_{\text{in}} \approx h_{FE} \cdot R_E$), minimizing circuit loading. PNG

4-2. Sketch the circuit of a practical (two-transistor) emitter-follower voltmeter. Carefully explain the circuit operation.

Operation: Uses two matched transistors (Q_1 and Q_2) in a balanced bridge setup to compensate for temperature variations and V_{BE} drift. Q_1 acts as the active emitter follower driven by input E, while Q_2 acts as a reference stage. A potentiometer (R_5) connected between emitters provides zero adjustment to balance bridge currents when $E = 0\ \text{V}$. PNG + 2

4-3. Using illustrations, discuss the problems that can occur with ground terminals when using an electronic voltmeter. Explain the need for a floating power supply in an electronic instrument.

Grounding Problem: If both the circuit under test and the electronic voltmeter share an Earth ground, connecting the voltmeter ground lead across a component whose terminals are both elevated above ground creates a short circuit through the ground path. PNG

Floating Power Supply: Using an isolated line transformer or battery power ("floating supply") isolates the instrument's circuit ground from Earth ground, enabling safe differential measurements across any component. PNG

4-4. Sketch a typical input attenuator as used with an electronic voltmeter. Show typical resistor values and range terminals, and explain attenuator operation.

Operation: A high-resistance potential divider consisting of series-connected precision resistors (e.g., $800\ \text{k}\Omega$, $100\ \text{k}\Omega$, $60\ \text{k}\Omega$, $40\ \text{k}\Omega$ total $1\ \text{M}\Omega$). The range selector taps off a specific fraction of input voltage so that maximum allowable voltage reaches the amplifier stage regardless of selected range. PNG + 2

4-5. Sketch the complete circuit of an emitter-follower voltmeter using a FET input stage. Carefully explain the circuit operation and the effect of the FET.

Operation & FET Effect: A JFET connected as a source follower is placed before the BJT emitter follower. Because a reverse-biased JFET gate draws negligible current ($I_G \approx 0$), the input resistance of the voltmeter increases to tens or hundreds of Megohms ($10\ \text{M}\Omega$ to $100\ \text{M}\Omega$), completely independent of the voltage range selected. PNG + 1

4-6. Draw a complete diagram of a transistor difference-amplifier type of voltmeter. Explain the circuit operation and discuss its advantage over emitter-follower voltmeters.

Operation & Advantage: Uses two transistors sharing a common emitter tail resistor R_E. Differential input voltage causes one collector voltage to drop and the other to rise; the PMMC meter connected between collectors measures this differential output. It provides higher voltage gain, excellent common-mode rejection, and superior thermal stability compared to basic emitter followers. PNG + 3

4-7. Sketch the complete circuit of an op-amp voltage follower voltmeter. Explain the circuit operation, and compare it to the simple emitter-follower voltmeter.

Operation: The input voltage is applied directly to the non-inverting input of an op-amp, with output shorted to the inverting input ($A_v = 1$). The PMMC meter circuit is connected to the op-amp output. PNG + 1

Comparison: Provides near-infinite input resistance, near-zero output resistance, and eliminates V_{BE} drop errors inherent in transistor emitter followers. PNG

4-8. Draw the circuit diagram of an op-amp noninverting amplifier voltmeter. Explain its operation, and compare it to the voltage-follower voltmeter.

Operation: The non-inverting amplifier provides closed-loop voltage gain $A_v = 1 + \frac{R_3}{R_4}$. This amplification allows full-scale meter deflection for extremely small input voltages (e.g., $10\ \mathrm{mV}$ to $100\ \mathrm{mV}$), unlike the voltage follower which requires a minimum input voltage equal to full-scale meter voltage. PNG + 1

4-9. Draw the circuit diagram of a voltmeter using an op-amp voltage-to-current converter. Explain the circuit operation.

Operation: The PMMC meter is placed directly in the feedback loop of the op-amp. The meter current $I_m = \frac{E_{\text{in}}}{R_3}$ is strictly controlled by the input voltage and feedback

resistor R_3, making meter indication completely independent of internal coil resistance R_m or diode voltage drops. PNG + 1

4-10 to 4-12. Electronic Ohmeters (Series, Shunt, and Linear)

Series Ohmmeter: Standard resistor and unknown R_x form a potential divider. Output voltage is non-linear relative to R_x. PNG + 1

Shunt Ohmmeter: Unknown R_x is shunted across the meter/input circuit, useful for low resistances. PNG

Linear Ohmmeter: Uses a constant-current source to pass fixed current I through R_x. Measured voltage $V_x = I \cdot R_x$ produces a perfectly **linear** resistance scale. PNG + 1

4-13 to 4-18. AC Electronic Voltmeters & Rectifiers

Precision Rectifiers: Utilize operational amplifiers in feedback loops to enclose rectifier diodes. This eliminates the $0.7\ \text{V}$ diode forward voltage drop threshold, permitting accurate AC measurements down to millivolt levels. PNG + 1

Numerical Problems Solutions

Problem 4-1

Given $V_{CC} = 12\ \text{V}$, $R_m = 1\ \text{k}\Omega$, $I_m = 2\ \text{mA}$, $h_{FE} = 80$, $E = 5\ \text{V}$ for FSD. Assumed $V_{BE} = 0.7\ \text{V}$. PNG + 1

$$\text{Emitter Voltage } V_E = E - V_{BE} = 5\ \text{V} - 0.7\ \text{V} = 4.3\ \text{V}$$

$$\text{Total Emitter Resistance } (R_s + R_m) = \frac{V_E}{I_m} = \frac{4.3\ \text{V}}{2\ \text{mA}} = 2.15\ \text{k}\Omega = 2150\ \Omega$$

$$\text{Multiplier Resistance } R_s = 2150\ \Omega - 1000\ \Omega = \mathbf{1150\ \Omega} \quad (\mathbf{1.15\ k\Omega})$$

$$\text{Input Resistance } R_{\text{in}} = (1 + h_{FE})(R_s + R_m) = 81 \times 2.15\ \text{k}\Omega = \mathbf{174.15\ k\Omega}$$

Problem 4-2

Given $V_{CC} = +9\ \text{V}, V_{EE} = -9\ \text{V}, R_1 = 12\ \text{k}\Omega, R_2 = R_3 = 2.7\ \text{k}\Omega, R_4 = R_6 = 3.3\ \text{k}\Omega, R_5 = 500\ \Omega, R_s + R_m = 10\ \text{k}\Omega, I_m = 100\ \mu\text{A}, h_{FE} = 75, E = 0\ \text{V}.$

PNG

Potential at wiper (V_p): $V_p = V_B - V_{BE} = 0\ \text{V} - 0.7\ \text{V} = \mathbf{-0.7\ V}$

Emitter Current (I_4):

$$I_4 = \frac{V_p - V_{EE}}{R_4} = \frac{-0.7\ \text{V} - (-9\ \text{V})}{3.3\ \text{k}\Omega} = \frac{8.3\ \text{V}}{3.3\ \text{k}\Omega} = \mathbf{2.515\ mA}$$

Base Currents (I_{B1}, I_{B2}):

$$I_{B1} = I_{B2} = \frac{I_4}{1 + h_{FE}} = \frac{2.515\ \text{mA}}{76} = \mathbf{33.09\ \mu A}$$

Current through R_2 (I_2):

$$I_2 = \frac{V_{CC} - V_{B1}}{R_2} = \frac{9\ \text{V} - 0\ \text{V}}{2.7\ \text{k}\Omega} = \mathbf{3.333\ mA}$$

Range of adjustment for V_p:

$$\Delta V_p = \pm I_4 \times \left(\frac{R_5}{2}\right) = \pm 2.515\ \text{mA} \times 250\ \Omega = \pm\mathbf{0.629\ V} \quad (\pm\mathbf{628.8\ mV}$$

Problem 4-3

Using circuit from Problem 4-2 ($R_s + R_m = 10\ \text{k}\Omega, I_{\text{FSD}} = 100\ \mu\text{A}$). PNG

$$\text{Meter Current } I_m = \frac{\Delta E}{R_s + R_m}$$

At $E = 0.6\ \text{V}$: $I_m = \frac{0.6\ \text{V}}{10\ \text{k}\Omega} = 60\ \mu\text{A} \implies \mathbf{60\%}\ \text{FSD} \quad (\mathbf{0.6}\ \text{FSD})$

At $E = 0.75\ \text{V}$: $I_m = \frac{0.75\ \text{V}}{10\ \text{k}\Omega} = 75\ \mu\text{A} \implies \mathbf{75\%}\ \text{FSD} \quad (\mathbf{0.75}\ \text{FSD})$

At $E = 1.0\ \text{V}$: $I_m = \frac{1.0\ \text{V}}{10\ \text{k}\Omega} = 100\ \mu\text{A} \implies \mathbf{100\%}\ \text{FSD} \quad (\mathbf{1.0}\ \text{FSD})$

Problem 4-5

Given $R_1 = 6.8\ \text{k}\Omega, R_2 = R_3 = 4.7\ \text{k}\Omega, R_4 = 1.5\ \text{k}\Omega, R_5 = 500\ \Omega, R_6 = 3.3\ \text{k}\Omega$, $R_s + R_m = 20\ \text{k}\Omega, I_{\text{FSD}} = 50\ \mu\text{A}, V_{CC} = +10\ \text{V}, V_{EE} = -10\ \text{V}, h_{FE} = 80$, $V_{GS} = -3\ \text{V}, E = 0\ \text{V}$. PNG

Potential V_p: $V_p = V_G - V_{GS} = 0\ \text{V} - (-3\ \text{V}) = +\mathbf{3.0}\ \text{V}$

Current I_1: $I_1 = \frac{V_{CC} - V_p}{R_1} = \frac{10\ \text{V} - 3\ \text{V}}{6.8\ \text{k}\Omega} = \mathbf{1.029}\ \text{mA}$

Current I_4: $I_4 = \frac{V_{CC} - V_p}{R_4} = \frac{10\ \text{V} - 3\ \text{V}}{1.5\ \text{k}\Omega} = \mathbf{4.667}\ \text{mA}$

Current I_6: $I_6 = \frac{V_E - V_{EE}}{R_6} = \frac{(3\ \text{V} - 0.7\ \text{V}) - (-10\ \text{V})}{3.3\ \text{k}\Omega} = \frac{12.3\ \text{V}}{3.3\ \text{k}\Omega} = \mathbf{3.727}\ \text{mA}$

Adjustment Range for V_p: $\Delta V_p = \pm I_4 \times 250\ \Omega = \pm\mathbf{1.167}\ \text{V}$

Problem 4-6

Attenuator set to $5\ \text{V}$ range ($5\ \text{V}$ input gives full scale deflection). PNG

At $1\ \text{V}$ input: Deflection $= \frac{1\ \text{V}}{5\ \text{V}} = \mathbf{20\%}\ \text{FSD} \quad (10\ \mu\text{A})$

At $3\ \text{V}$ input: Deflection $= \frac{3\ \text{V}}{5\ \text{V}} = \mathbf{60\%}\ \text{FSD} \quad (30\ \mu\text{A})$

At $4\ \text{V}$ input: Deflection $= \frac{4\ \text{V}}{5\ \text{V}} = \mathbf{80\%}\ \text{FSD} \quad (40\ \mu\text{A})$

Problem 4-7

Given $R_1 = R_2 = 15\ \text{k}\Omega$, $R_{L1} = R_{L2} = 3.9\ \text{k}\Omega$, $R_E = 3.3\ \text{k}\Omega$, $R_s = 33\ \text{k}\Omega$, $R_m = 750\ \Omega$, $V_{CC} = +12\ \text{V}$, $V_{EE} = -12\ \text{V}$, $E = 0\ \text{V}$. PNG

Common Emitter Voltage (V_E): $V_E = 0\ \text{V} - 0.7\ \text{V} = \mathbf{-0.7}\ \text{V}$

Total Tail Current (I_E): $I_E = \frac{-0.7-(-12)}{3.3\ \text{k}\Omega} = 3.424\ \text{mA}$

Transistor Emitter Currents: $I_{E1} = I_{E2} = \frac{3.424\ \text{mA}}{2} = \mathbf{1.712}\ \text{mA}$

Collector Voltages (V_{C1}, V_{C2}):

$$V_{C1} = V_{C2} = V_{CC} - I_C R_{L1} = 12\ \text{V} - (1.712\ \text{mA} \times 3.9\ \text{k}\Omega) = \mathbf{5.323}\ \text{V}$$

Problem 4-8

Given $h_{FE} = 100$, $h_{ie} = 1.2\ \text{k}\Omega$, $I_{\text{FSD}} = 50\ \mu\text{A}$. PNG

Differential Voltage Gain (A_d):

$$A_d = \frac{h_{fe} R_L}{2h_{ie}} = \frac{100 \times 3.9\ \text{k}\Omega}{2 \times 1.2\ \text{k}\Omega} = 162.5$$

Differential Output Voltage for FSD:

$$V_{\text{diff}} = I_m(R_s + R_m + 2R_L) = 50\ \mu\text{A} \times (33\ \text{k}\Omega + 0.75\ \text{k}\Omega + 7.8\ \text{k}\Omega) = 2.0$$

Required Input Voltage (E):

$$E = \frac{V_{\text{diff}}}{A_d} = \frac{2.0775\ \text{V}}{162.5} = \mathbf{0.01278}\ \text{V} \quad (\mathbf{12.78}\ \text{mV})$$

Problem 4-11

Given $I_{\text{FSD}} = 37.5\ \mu\text{A}$, $R_3 = 80\ \text{k}\Omega$. PNG

$$E = I_m \cdot R_3$$

For FSD ($37.5\ \mu\text{A}$): $E = 37.5\ \mu\text{A} \times 80\ \text{k}\Omega = \mathbf{3.0}\ \text{V}$

For 0.5 FSD ($18.75\ \mu\text{A}$): $E = 0.5 \times 3.0\ \text{V} = \mathbf{1.5}\ \text{V}$

Problem 4-12

Given $E = 1.0\ \text{V}$ for FSD ($I_{\text{FSD}} = 37.5\ \mu\text{A}$), $R_m = 900\ \Omega$. PNG

New R_3:

$$R_3 = \frac{1.0\ \text{V}}{37.5\ \mu\text{A}} = \mathbf{26.67\ \text{k}\Omega}$$

Op-amp Output Voltage (V_o):

$$V_o = E + I_m R_m = 1.0\ \text{V} + (37.5\ \mu\text{A} \times 900\ \Omega) = \mathbf{1.0338}\ \text{V}$$

Problem 4-13

In a series electronic ohmmeter circuit (Fig 4-9):

PNG

$$\text{Deflection Fraction } s = \frac{R_x}{R_x + R_{\text{std}}}$$

At 25% FSD ($s = 0.25$): $R_x = \frac{0.25}{0.75} R_{\text{std}} = \mathbf{\frac{1}{3} R_{\text{std}}}$

At 75% FSD ($s = 0.75$): $R_x = \frac{0.75}{0.25} R_{\text{std}} = \mathbf{3 R_{\text{std}}}$

Problem 4-14

Given $R_{\text{std}} = 100\ \text{k}\Omega$, $R_x = 166\ \text{k}\Omega$. PNG

$$\%\ \text{Deflection} = \left(\frac{166\ \text{k}\Omega}{166\ \text{k}\Omega + 100\ \text{k}\Omega}\right) \times 100\% = \frac{166}{266} \times 100\% = \mathbf{62.4\%}\ \text{FSD}$$

Problem 4-17

Given $I_{\text{FSD}} = 500\ \mu\text{A}$, $R_m = 460\ \Omega$, $R_s = 450\ \Omega$. PNG

Total meter circuit resistance $R_{\text{total}} = 450 + 460 = 910\ \Omega$.
For half-wave average rectified current:
PNG

$$V_{\text{rms}} = 2.2214 \times I_m \times R_{\text{total}} = 2.2214 \times (500\ \mu\text{A}) \times 910\ \Omega = \mathbf{1.011}\ \text{V}$$

Problem 4-18

Using precision half-wave rectifier with $R_3 = R_s = 450\ \Omega$, $I_{\text{FSD}} = 500\ \mu\text{A}$: PNG

RMS voltage for FSD:

$$V_{\text{rms}} = 2.2214 \times (500\ \mu\text{A}) \times 450\ \Omega = \mathbf{0.50}\ \text{V} \quad (\mathbf{500}\ \text{mV})$$

At $100\ \text{mV}$ input: Deflection $= \frac{100\ \text{mV}}{500\ \text{mV}} = \mathbf{20\%}\ \text{FSD}$

At $200\ \text{mV}$ input: Deflection $= \frac{200\ \text{mV}}{500\ \text{mV}} = \mathbf{40\%}\ \text{FSD}$

Problem 4-19

Given $R_1 = 22\ \text{k}\Omega$, $R_2 = 2.25\ \text{k}\Omega$, $R_3 = 6.8\ \text{k}\Omega$, $R_s + R_m = 1\ \text{k}\Omega$, $I_{\text{FSD}} = 300\ \mu\text{A}$. PNG

Amplifier Voltage Gain (A_v):

$$A_v = 1 + \frac{R_1}{R_2} = 1 + \frac{22\ \text{k}\Omega}{2.25\ \text{k}\Omega} = 10.778$$

Input RMS Voltage for FSD:

$$V_{\text{rms}} = \frac{2.2214 \times (300\ \mu\text{A}) \times 1\ \text{k}\Omega}{10.778} = \mathbf{61.83}\ \text{mV}$$

Input RMS Voltage for 0.5 FSD:

$$V_{\text{rms}} = 0.5 \times 61.83\ \text{mV} = \mathbf{30.92}\ \text{mV}$$

Problem 4-20

PNG + 4

Full-wave precision rectifier with $I_{\text{FSD}} = 500\ \mu\text{A}$, $R_3 = 450\ \Omega$. PNG

For full-wave rectified output:

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$$V_{\text{rms}} = 1.1107 \times I_m \times R_3 = 1.1107 \times (500\ \mu\text{A}) \times 450\ \Omega = \mathbf{0.25}\ \text{V} \quad (\mathbf{250}\ \text{mV})$$